Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/08/2009
PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21224-6823	**Congressional District:**	3
Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Hursh, Steven (Institutes for Behavior Resources, Inc.)		
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The aims of this project focus upon the development of an experimental test bed for modeling performance effectiveness and psychosocial adaptation in support of exploratory spaceflight missions beyond Earth's atmosphere. The research methodology employed provides for distributed interactive multi-person simulations in computer-generated environments focused upon psychosocial performance interactions within and between space-dwelling flight crews and Earth-based control center groups. The simulation provides an automated means for setting the context for the analysis of space crew performance as well as monitoring electronically the interactive effects of simulated communication modality constraints, mission management systems, and other stressful conditions. Within this context, the objectives of this project are to provide risk assessment and countermeasure evaluation of the following fundamental behavioral interaction operations that will most likely affect crew performance effectiveness and psychosocial adaptation: 1) the structure and function of communication channels within and between simulated space-dwelling and Earth-based groups; 2) factors associated with variations in the behavioral management systems between space-dwelling and

Task Description:	Earth-based groups; 3) factors associated with variations in workloads, stressful time pressure, and conflict conditions; and 4) behavioral and psychosocial interaction systems between spaceflight crews and Earth-centered mission support operations that are most likely to influence individual and group performance during long-duration missions. The results of these distributed interactive simulation studies have shown clearly that cooperative and productive interactions are maintained between individually isolated and dispersed crewmembers in a task-driven environment, and that experimental flight crews actively engage in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, investigations of communication modality constraints have shown a high degree of interchangeability between available communication modes that acted as a countermeasure to maintain effective crew performance. Further, variations in crew configurations have shown that performance effectiveness levels are significantly reduced during missions in which crew assignment changes resulted in inexperienced crew configurations, as contrasted with highly-experienced crew configurations. Studies of positive and negative incentive conditions on crew performance effectiveness have also shown increased performance effectiveness under both positive and negative incentive conditions, but an adverse change in psychosocial adaptation. Thus incentive conditions associated with simulated spaceflight missions can significantly affect psychosocial adaptation without compromising task performance effectiveness in trained and experienced volunteer crews. Newly-initiated studies on management system variations consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Tests of these methods revealed clear benefits of autonomy that improved crew performance and decreased negative psychosocial adaptation self-report measures. Autonomous missions also decreased undirected linguistic expressions of negative emotion but increased expressions of social processes and achievement. Salivary cortisol (an index of physiological stress reactivity) was lower during autonomous versus scheduled missions, and cortisol reactivity during autonomous missions predicted subjective responses to the loss of autonomy. The psychosocial and performance effects of autonomy were generally robust to communications constraints. Assessment of crew performance and psychosocial adaptation has also been extended to longer-duration, 12-hour mission simulations conducted during different parts of a 24-hour circadian cycle (either 'Early': 9 am-9 pm, or 'Late': 9 pm-9 am). 'Late' workloads produced significant decrements in individual performances that included reaction time deficits, attentional lapses, psychosocial communication reductions, failures to properly report rule solutions, inadvertent discarding of high-valued geologic samples, and significant psychosocial changes. Importantly, in spite of these numerous decrements, crew performance efficiency remained normal. Lapses in individual performances thus appeared to be compensated for in the overall crew performance, indicating the utility of multi-person crews as countermeasures to reductions in individual performances under adverse workload conditions. Nevertheless, the dramatic decrease in psychosocial communication during late missions could be an early warning sign of crew desynchronization that could hinder crew cohesion and adaptability to malfunctions, environmental disruptions, or schedule demands. Design and development was also initiated for a rapid, simple, and objective diagnostic assessment of crew cohesion - the Team Performance Task (TPT). Results from laboratory tests of several TPT prototypes support the general paradigm's sensitivity to individual differences in pro-social behavior in an interdependent group context, a fundamental component of group cohesion. Future plans involve extended mission management and mission workload studies, migration of the simulation scenario to internet-based operations, and scenario modifications to provide for a broader range of surface regions to enhance the complexity of geologic identification, collection and conceptual analysis task. Variations in core communication functions under this more complex model will continue to focus upon an experimental analysis of the interactions between ground-based control and flight crew performance effectiveness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The methodological development associated with this NSBRI-funded research has provided a test bed for modeling performance effectiveness and psychosocial adaptation in computer-generated distributed interactive multi-person environments. This experimental setting provides an automated means for setting the task performance context as well as monitoring electronically the interactive effects of communication modality constraints and other environmentally stressful conditions. Such a distributed interactive experimental analysis can characterize the extent to which variations in the structure and function of communication channels within and between individuals and groups affect the ability to accurately discern and respond to the realities and requirements of both operational and interpersonal transactions. The effects of environmentally stressful time constraints, appetitive and aversive characteristics of programmed incentive conditions, workload conditions, and management system conditions can be analyzed experimentally in the context of such distributed interactive environments. Research conducted within the context of this distributed interactive simulation model can provide the basis for developing effective patterns of communication and problem solving strategies as well as a range of training procedures to enhance problem solving effectiveness. This project's research on better ways to assess, train and manage small team performance effectiveness under hazardous and stressful conditions is relevant to transportation agencies, military forces, and first responders. The Earth benefits to be derived from the research extend to small operational group selection and training procedures, to the management of stressful interactions, and to the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded knowledge base.

Task Progress:	Task progress included newly-initiated studies on the effects of management system variations and workload condition variations undertaken during the current reporting year. The newly-initiated studies on management system variations consisted of developing methodologies for assessing crew performance effectiveness under rigid schedule-based management of crew activities by Mission Control versus more flexible, contingency-based, autonomous management of crew activities by themselves. Tests of these methods revealed clear benefits of autonomy resulting in improved performance and decreased negative psychosocial and biobehavioral adaptation measures. Studies during the current reporting period have extended the experimental analyses described above to include variations in communication constraints under differing crew management conditions. Early results support the beneficial effects of autonomy, and suggest that these effects are robust to communication constraints. Taken together, these experiments reveal that crew autonomy during simulated spaceflight missions can positively affect team performance effectiveness and individual self-report measures of morale. Mission goals and behavioral health objectives can be satisfied, and even enhanced, under autonomous management conditions in well-trained and cohesive crews. Major software architecture changes have been made to improve and expand features of the geologic simulation. Due to the older software's developmental and support limitations, the simulation has been rewritten in a more optimal, industry standard development environment (ActionScript, Adobe Flash) that has speeded and simplified development, and now affords more flexibility in simulation design, higher quality simulation production, greater ease of software change for future growth, and the capability to operate transparently over local networks or over the Internet. Major functional enhancements of the software include: separation of 'Mission Control' and 'Flight Operations' roles, increased numbers of vehicles participating in scenarios (i.e., increased team sizes), and the addition of actual lunar surface backgrounds. Crews have already successfully interacted with the new simulation software, including successful 3-person interactions via the internet from remote locations (e.g., individuals located separately at IBR, The Johns Hopkins Medical School, and at the University of Maryland, Baltimore County). Conceptual development of the Team Performance Task (TPT) has undergone several prototype designs and programming refinements, and data collection aimed at building an empirical foundation for defining the task's parameters has begun. Experimental assessments of Individual vs. Collective reward schemes and Anonymity vs. Identity will follow alpha testing, as will a comprehensive validation process in established cohesive and non-cohesive groups.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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